5 Kvadrat tenglamalar I — ko'paytuvchalash

1-to'plam. *Ko'paytuvchalarga ajrating: $a = 1$, b va c musbat.*

1. $x^2 + 5x + 4 = 0$

2. $x^2 + 6x + 8 = 0$

3. $x^2 + 9x + 14 = 0$

4. $x^2 + 3x + 2 = 0$

5. $x^2 + 10x + 16 = 0$

6. $x^2 + 8x + 12 = 0$

7. $x^2 + 11x + 18 = 0$

8. $x^2 + 7x + 10 = 0$

2-to'plam. *Ko'paytuvchalarga ajrating: manfiy b va/yoki c.*

1. $x^2 - 5x + 6 = 0$

2. $x^2 + 2x - 15 = 0$

3. $x^2 - 3x - 10 = 0$

4. $x^2 - 7x + 12 = 0$

5. $x^2 + 4x - 21 = 0$

6. $x^2 - 2x - 24 = 0$

7. $x^2 - x - 6 = 0$

8. $x^2 + x - 20 = 0$

3-to'plam. *Avval umumiy ko'paytuvchini chiqaring, so'ng $a \neq 1$ holatini ko'paytuvchalang.*

1. $3x^2 + 9x + 6 = 0$

2. $2x^2 - 8 = 0$

3. $2x^2 + 5x + 2 = 0$

4. $3x^2 - x - 2 = 0$

5. $4x^2 + 4x - 8 = 0$

6. $2x^2 - 3x - 5 = 0$

7. $5x^2 + 15x + 10 = 0$

8. $2x^2 + 7x + 3 = 0$

4-to'plam. *Avval hammasini nolga tenglashtiring, so'ng yeching.*

1. $(x + 2)(x - 1) = 4$

2. $(x - 2)(x + 5) = 18$

3. $(x + 1)(x + 4) = 10$

4. $(x - 3)(x - 2) = 2$

5. $(x + 6)(x - 1) = -10$

6. $(x - 4)(x + 3) = -6$

5-to'plam. *Tenglamani yeching, so'ng savolda aniq so'ralgan narsaga javob bering.*

1 $x^2 - 2x - 15 = 0$. Ildizlar **yig'indisini** toping: . . . . . .

2 $x^2 + x - 12 = 0$. Ildizlar **ko'paytmasini** toping: . . . . . .

3 $x^2 - 7x + 6 = 0$. Ildizlar orasidagi **musbat ayirmani** toping (katta minus kichik): . . . . . .

4 $x^2 + 3x - 10 = 0$. Ildizlar **yig'indisini** toping: . . . . . .

6-to'plam. *Xatoni toping – buzilgan qadamni ayting, to'liq to'g'ri javobni bering (ikkala ildiz, agar ikkita bo'lsa).*

1 $(x+3)(x-2) = 0$ ifodasida ko'rsatilgan yechimda xato bor:

"Yechish: $x + 3 = 0$, *shuning uchun* $x = -3$."

Xato va to'g'ri javob: . . . . . .

2 $x^2 - 4x - 5 = 0$ ifodasida ko'rsatilgan yechimda xato bor:

"Yechish: $(x-1)(x+5) = 0$, *shuning uchun* $x = 1$ *yoki* $x = -5$."

Xato va to'g'ri javob: . . . . . .

3 $(x-2)(x+4) = 16$ ifodasida ko'rsatilgan yechimda xato bor:

"Yechish: $x - 2 = 16$ *yoki* $x + 4 = 16$, *shuning uchun* $x = 18$ *yoki* $x = 12$."

Xato va to'g'ri javob: . . . . . .

4 $2x^2 + 6x = 0$ ifodasida ko'rsatilgan yechimda xato bor:

"Yechish: $2x^2 = -6x$, *ikkala tomonni* x *ga bo'lamiz:* $2x = -6$, $x = -3$."

Xato va to'g'ri javob: . . . . . .

7-to'plam. *Bonus / qiyin.*

1. $6x^2 + 5x + 1 = 0$

2. $4x^2 - 9 = 0$

3. $x^2 - \frac{1}{4} = 0$

4. $x^2 - 3x + 2 = 12$

5. $2x^2 + x - 3 = 7$

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$-4, -1$	2	$-4, -2$
	3	$-7, -2$	4	$-2, -1$
	5	$-8, -2$	6	$-6, -2$
	7	$-9, -2$	8	$-5, -2$
2-to'plam	1	$2, 3$	2	$-5, 3$
	3	$-2, 5$	4	$3, 4$
	5	$-7, 3$	6	$-4, 6$
	7	$-2, 3$	8	$-5, 4$
3-to'plam	1	$-2, -1$	2	$-2, 2$
	3	$-\frac{1}{2}, -2$	4	$-\frac{2}{3}, 1$
	5	$-2, 1$	6	$-1, \frac{5}{2}$
	7	$-2, -1$	8	$-\frac{1}{2}, -3$
4-to'plam	1	$-3, 2$	2	$-7, 4$
	3	$-6, 1$	4	$1, 4$
	5	$-4, -1$	6	$-2, 3$
5-to'plam	1	2	2	-12
	3	5	4	-3
7-to'plam	1	$-\frac{1}{2}, -\frac{1}{3}$	2	$-\frac{3}{2}, \frac{3}{2}$
	3	$-\frac{1}{2}, \frac{1}{2}$	4	$-2, 5$
	5	$-\frac{5}{2}, 2$	–	–
6-to'plam	1. Xato – bitta ildizda to'xtagan. To'g'ri: $x = -3$ yoki $x = 2$ (ikkalasi).			
	2. Xato – noto'g'ri ishora juftligi. To'g'ri: $(x-5)(x+1) = 0$, $x = 5$ yoki $x = -1$.			
	3. Xato – o'ng tomon 0 emas. To'g'ri: $x^2 + 2x - 24 = 0 \Rightarrow (x+6)(x-4) = 0 \Rightarrow x = -6$ yoki $x = 4$.			
	4. Xato – x ga bo'lish $x = 0$ yechimini yo'qotadi. To'g'ri: $2x(x+3) = 0 \Rightarrow x = 0$ yoki $x = -3$.			

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

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